Parametrizing by the Ellentuck space

Timothy Trujillo

COLORADO SCHOOL OF MINES
EARTH • ENERGY • ENVIRONMENT

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Overview

- Topological Ramsey spaces and the abstract Ellentuck theorem
- The Ellentuck and Milliken spaces
- Parametrizing by a topological Ramsey space
- The main result and its corollaries
- Application: n-dimensional parametrized perfect set theorem

The Ellentuck Topology

Definition

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The **Ellentuck topology** on $\mathcal{R}$ is the topology generated by the sets $[a, B]$ where a is in the range of r and $B \in \mathcal{R}$.

Topological Ramsey Spaces

Definition

- A subset $\mathcal{X}$ of $\mathcal{R}$ is **Ramsey** if for every $\emptyset \neq [a, A]$, there is a $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \cap \mathcal{X} = \emptyset$.

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- A subset $\mathcal{X}$ of $\mathcal{R}$ is **Ramsey null** if for every $\emptyset \neq [a, A]$, there is a $B \in [a, A]$ such that $[a, B] \cap \mathcal{X} = \emptyset$.

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- For $n < \omega$ and $X = \{x_0, x_1, \dots\}$ in $[\omega]^\omega$ enumerated in increasing order $r(X, n) = \{x_0, \dots, x_{n-1}\}$.
- The **Ellentuck space** is the topological Ramsey space associated to the triple $([\omega]^\omega, \subseteq, r)$.

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- FIN^n is the collection of all block sequences of length n.
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- If X is a block sequence then $FU(X)$ denotes the subset of FIN consisting of all finite unions of the elements of the sequence X.

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Theorem (Milliken, [2])

The triple $(\mathrm{FIN}^\infty, \leq, r)$ *is a topological Ramsey space.*

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Theorem (Todorcevic, [1])

If $(\mathcal{R}, \leq, r)$ is a closed subspace of $(\mathrm{ran}(r))^\omega$ and satisfies **A.1-A.4** *then $(\mathcal{R}, \leq, r)$ is a topological Ramsey space.*

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$$r(X, Y, n) = \begin{cases} \emptyset & \text{if } n = 0, \\ (r(X, n), r(Y, n)) & \text{otherwise.} \end{cases}$$

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We say that $\mathcal{R}$ **parametrizes** $\mathcal{S}$ if there is a quasi-order $\leq$ on $\mathcal{R} \times \mathcal{S}$ such that $(\mathcal{R} \times \mathcal{S}, \leq, r)$ is a topological Ramsey space.

Main Result

Theorem (T. [3])

Suppose $(\mathcal{R}, \leq, r)$ *and* $(\mathcal{S}, \leq, r)$ *are closed subspaces of* $\mathcal{AR}^\omega$ *and* $\mathcal{AS}^\omega$*, respectively, and satisfy* **A.1**-**A.4**. *Let* $\leq$ *be a quasi-order on* $\mathcal{R} \times \mathcal{S}$ *such that* $(\mathcal{R} \times \mathcal{S}, \leq, r)$ *satisfies* **A.1**-**A.3**. *If* $\mathcal{R}$ *'diagonalizes'* $\mathcal{S}$ *with respect to the quasi-order* $\leq$ *on* $\mathcal{R} \times \mathcal{S}$ *then* $\mathcal{R}$ *is parametrized by* $\mathcal{S}$. *In particular,* $(\mathcal{R} \times \mathcal{S}, \leq, r)$ *is a topological Ramsey space that satisfies* **A.1**-**A.4**.

Corollaries

Corollary (T. [3])

Let $(\mathcal{R}, \leq, r)$ *be a closed subspace of* $\mathrm{ran}(r)^\omega$ *satisfying* **A.1**-**A.4**. *If for all* s, t *in the range of* r, $s \sqsubseteq t \implies s \leq_{\mathrm{fin}} t$, *then the topological Ramsey space* $(\mathcal{R}, \leq, r)$ *parametrizes the Ellentuck space.*

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Corollary (T. [3])

The Ellentuck space parametrizes itself. That is, there is a quasi-order $\ll$ *on* $[\omega]^\omega \times [\omega]^\omega$ *such that* $([\omega]^\omega \times [\omega]^\omega, \ll, r)$ *is a topological Ramsey space.*

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Corollary (T. [3])

The Milliken space parametrizes the Ellentuck space. That is, there is a quasi-order $\ll$ *on* $\mathrm{FIN}^\infty \times [\omega]^\omega$ *such that* $(\mathrm{FIN}^\infty \times [\omega]^\omega, \ll, r)$ *is a topological Ramsey space.*

Depth

Remark

Axiom **A.2** (Finitization) requires the existence of a quasi-order $\leq_{\mathrm{fin}}$ on $\mathrm{ran}(r)$. If **A.2** holds, $X \in \mathcal{R}$ and $a \in \mathrm{ran}(r)$ then we let

$$\mathrm{depth}_X a = \begin{cases} \min\{n \in \omega : a \leq_{\mathrm{fin}} r(X, n)\} & \text{if } \exists n, a \leq_{\mathrm{fin}} r(X, n) \\ \infty & \text{otherwise.} \end{cases}$$

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Remark

The Ellentuck space does not parametrize the Milliken space via the quasi-order $\ll$.

Parametrized perfect set theorem

Theorem (parametrized perfect set theorem, [1])

For every Souslin-measurable coloring of $\mathbb{R} \times [\omega]^\omega$ with finitely many colors there exists a perfect set $P \subseteq \mathbb{R}$ and infinite set $Y \subseteq \omega$ such that the coloring is monochromatic on $P \times [Y]^\omega$.

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Theorem (n-dimensional parametrized perfect set theorem, [1])

Let $0 < n < \omega$. For every Souslin-measurable coloring of $\mathbb{R}^n \times [\omega]^\omega$ with finitely many colors there exists a sequence $(P_i)_{i<n}$ of perfect sets subsets of $\mathbb{R}$ and an infinite set $Y \subseteq \omega$ such that the coloring is monochromatic on $\prod_{i=0}^{n-1} P_i \times [Y]^\omega$.

Proof Sketch

Since we are constructing a perfect set on the first coordinate we can work with the subset of irrationals which is homeomorphic to $[\omega]^\omega$ with its metric topology. Without loss of generality it is enough to prove the statement for Souslin-measurable colorings of $[\omega]^\omega \times [\omega]^\omega$ with respect to the product topology where each factor is taken with its metric topology.

Let $c : [\omega]^\omega \times [\omega]^\omega \to k$ be a Souslin-measurable coloring with $k \in \omega$. For each $i < k$, $c^{-1}\{i\}$ is an analytic subset of $[\omega]^\omega \times [\omega]^\omega$. Since the collection of sets with the Baire property with respect to Ellentuck topology on $[\omega]^\omega \times [\omega]^\omega$ is closed under the Souslin operation each $c^{-1}\{i\}$ has the Baire property with respect to the Ellentuck topology. Since $([\omega]^\omega \times [\omega]^\omega, \leq, r)$ is a topological Ramsey space each $c^{-1}\{i\}$ is Ramsey. Thus there exists $(X, Y) \in [(\emptyset, \emptyset), (\omega, \omega)]$ such that for each $i < k$, either $[(\emptyset, \emptyset), (X, Y)] \subseteq c^{-1}\{i\}$ or $[(\emptyset, \emptyset), (X, Y)] \cap c^{-1}\{i\} = \emptyset$. That is, c is monochromatic on $[(\emptyset, \emptyset), (X, Y)]$.

Thank You!

[1] Todorcevic, Introduction to Ramsey spaces, 2010.

[2] Milliken, Ramsey's theorem with sums or unions, J. Combinatorial Theory Ser. A 1975.

[3] Trujillo, Parametrizing by the Ellentuck space, preprint 2016.